

Notes: Chen and Wang (RFS, 2025)

Dynamic Market Making with Asymmetric Information and Market Power

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1 Big picture

- **Question.** Why do trading volume and bid-ask spreads often display **double U-shaped intraday patterns**? Can one model explain both patterns at the same time?
- **Setting.** The paper builds a multiperiod trading model with informed investors, uninformed investors, and **oligopolistic market makers**. Traders can trade repeatedly during the day, private information and liquidity shocks arrive over time, and market makers may have market power.
- **Core challenge.** Existing theories can explain either part of the intraday pattern, or they rely on exogenous order-flow assumptions. They do not give a unified endogenous explanation for why both **volume** and **spread** are high at the open and close.
- **Main conceptual contribution.** The paper combines **dynamic asymmetric information**, **liquidity-driven hedging demand**, and **imperfect competition among liquidity suppliers**. Market power makes nonstrategic traders smooth their trades over time.
- **Two core trading motives.**
 - **Information trading:** traders want to act early on private signals before others do.
 - **Hedging trading:** traders prefer to postpone trades linked to liquidity shocks because adverse selection becomes weaker later in the day.
- **Two market-making regimes.**
 - **Agency trading market makers:** they offset trades each round and do not carry inventory across time.
 - **Principal trading market makers:** they carry inventory and maximize total profits over the whole trading horizon.
- **Main equilibrium result.** Under agency trading, both spread and trading volume are proportional to the gap in reservation prices between informed and uninformed traders. Under principal trading, the market makers' own reservation price also matters.
- **Main dynamics result.** When information and liquidity shocks both arrive at constant rates, traders act quickly on information and delay hedging. As a result, both volume and bid-ask spreads can become **U-shaped**.
- **Key difference across market-making regimes.** With new liquidity shocks, agency market makers generate a **hump-shaped** volume spike before the close, because traders optimally shift some orders earlier to avoid a very wide late-day spread. Principal market makers instead generate volume that rises more smoothly toward the close.
- **Calibration result.** Calibrating the model to **NASDAQ TAQ data from 2000–2021** produces spread and turnover paths that line up reasonably well with the empirical intraday patterns. The calibrated parameters imply **more informative private signals at the open** and **stronger hedging uncertainty near the close**.
- **Empirical implications.** The model predicts that U-shaped spreads should be stronger when liquidity suppliers are less competitive or order imbalance is larger, and that the U-shape in

volume should be stronger when information asymmetry is lower or liquidity-trader activity is higher.

- **Economic takeaway.** The open and the close look special not because order flow is assumed to be special by construction, but because traders' incentives and market makers' pricing power interact dynamically over the day.

2 Model environment and key objects

2.1 Agents, assets, and shocks

There are three types of investors:

- **Informed investors**, who receive private information and also face hedging needs.
- **Uninformed investors**, who do not observe the private signal.
- **Market makers**, who are also uninformed but may have market power.

All traders can hold a risk-free asset and one risky security. The risky asset pays

$$V \sim N(\bar{V}, \tau_V^{-1}). \quad (1)$$

At each trading date $t = 0, 1, \dots, T$, informed investors observe a private signal

$$v_t = V + \varepsilon_t, \quad (2)$$

where ε_t is Gaussian noise with precision $\tau_{\varepsilon,t}$.

Informed traders also receive a nontradable risky endowment that creates a hedging motive:

$$X_t = X_{t-1} + \eta_t. \quad (3)$$

The nontradable asset pays $V - \bar{V}$ at date $T + 1$, so a larger X_t creates demand to hedge exposure to the same fundamental risk.

Interpretation. The model gives informed traders **two motives** to trade: they speculate on private information and they hedge a correlated nontradable exposure. That second motive is what allows the paper to generate heavy trading near the close.

2.2 Bid, ask, and market clearing

At each period, traders can sell to market makers at the bid price B_t , buy from them at the ask price A_t , or choose not to trade. Market makers choose bid and ask quantities simultaneously. Let α_t^j be the quantity sold by market maker j at the ask, and β_t^j the quantity bought by market maker j at the bid.

The bid and ask are pinned down by market clearing. Intuitively:

- total ask-side quantity supplied by market makers must equal total buy orders of other traders;

- total bid-side quantity demanded by market makers must equal total sell orders of other traders.

Interpretation. Market makers are not passive quote posters. They internalize the demand and supply schedules they face, and their chosen depths determine the prices needed to clear the market.

2.3 Information sets and the composite signal

The information sets are

$$\mathcal{F}_t^I = \{v_t, X_t, A_t, B_t\}, \quad \mathcal{F}_t^U = \mathcal{F}_t^M = \{A_t, B_t\}. \quad (4)$$

Although uninformed traders do not observe v_t or X_t , prices reveal a **composite signal**

$$s_t := v_t - h_t \eta_t = V + \varepsilon_t - h_t \eta_t. \quad (5)$$

Thus the market cannot separately infer information shocks and liquidity shocks from prices; it sees only their combination.

Kalman filtering then gives recursive updating rules for informed and uninformed beliefs. The key sufficient-statistic objects are the informed estimate $\hat{V}_t^I$, the uninformed estimate $\hat{V}_t^U$, and the implied signal-to-noise objects

$$\mu_t = \lambda_I o_{V,t}^I, \quad h_t = \lambda_I / \tau_{\varepsilon,t}. \quad (6)$$

Interpretation. Prices reveal some information over time, but not enough to fully separate whether a given trade is driven by superior information or by hedging needs.

2.4 Agency versus principal market makers

The paper studies two tractable special cases.

Agency trading market makers. Let $\lambda_M \rightarrow \infty$ and assume zero initial inventory. Then market makers optimally offset buy and sell trades each round and solve

$$\max_{\alpha_t^j = \beta_t^j} \mathbb{E}[\alpha_t^j A_t - \beta_t^j B_t \mid \mathcal{F}_t^M]. \quad (7)$$

They behave like spread-earning intermediaries with no intertemporal inventory management.

Principal trading market makers. Let $\lambda_M = \lambda_U = 0$. Then market makers carry inventory and solve

$$\max_{\alpha_t^j, \beta_t^j} \mathbb{E} \left[W_{-1}^j + \sum_{t=0}^T (\alpha_t^j A_t - \beta_t^j B_t) + \theta_T^j V \mid \mathcal{F}_t^M \right]. \quad (8)$$

Interpretation. The agency case isolates pure spread-setting with round-by-round offsetting. The principal case adds inventory management and makes the market makers' own valuation enter the equilibrium prices.

2.5 Calibration targets and stylized facts

The paper calibrates the model to intraday NASDAQ data. The key empirical objects are:

- **Absolute spread:** $\text{Ask}_{it} - \text{Bid}_{it}$.
- **Relative spread:** $(\text{Ask}_{it} - \text{Bid}_{it})/\text{Midpoint}_{it}$.
- **Turnover:** trading volume divided by shares outstanding over each half-hour interval.

The basic empirical patterns are strongly U-shaped:

- the average absolute spread is about 0.552 at 9:30, falls to roughly 0.104–0.112 around midday, and rises again to 0.299 at 16:00;
- turnover starts at 25.13% in the first half-hour, falls to about 10.05% around 13:30–14:00, and rises again to 18.79% in the last half-hour.

Interpretation. The empirical target is not just the level of spreads or trading volume. It is the **joint shape** of both objects over the day.

3 Models and equilibrium (all equations + interpretation)

3.1 Investor demand and reservation prices

Given bid and ask prices, the optimal demand schedule of type $i \in \{I, U\}$ in the agency case is piecewise linear:

$$\theta_t^i - \theta_{t-1}^i = \begin{cases} \frac{P_t^{iR} - A_t}{\gamma_t^i}, & A_t < P_t^{iR}, \\ 0, & B_t \leq P_t^{iR} \leq A_t, \\ -\frac{B_t - P_t^{iR}}{\gamma_t^i}, & B_t > P_t^{iR}. \end{cases} \quad (9)$$

Reservation prices summarize each trader's valuation of one extra unit of the risky asset. For the agency case they are

$$P_t^{IR} = \bar{V} + \xi_t^I(\hat{V}_t^I - \mu_t X_t - \bar{V}) + \psi_t^I(\hat{V}_t^U - \bar{V}) - f_t^I \bar{\theta} - \gamma_t^I \theta_{t-1}^I, \quad (10)$$

$$P_t^{UR} = \bar{V} + \xi_t^U(\hat{V}_t^I - \mu_t X_t - \bar{V}) + \psi_t^U(\hat{V}_t^U - \bar{V}) - f_t^U \bar{\theta} - \gamma_t^U \theta_{t-1}^U. \quad (11)$$

Define the reservation-price gap

$$\Delta_t := P_t^{IR} - P_t^{UR}. \quad (12)$$

The paper decomposes Δ_t into three pieces:

- an **estimated payoff difference**;
- a **hedging difference**;
- a **risk-premium difference**.

Interpretation. Δ_t is the central state variable for the agency equilibrium. When informed and uninformed valuations drift apart, both volume and the spread rise.

At the final trading date, the reservation prices simplify to

$$P_T^{IR} = \hat{V}_T^I - \mu_T X_T - \lambda_I o_{V,T}^I \theta_{T-1}^I, \quad P_T^{UR} = \hat{V}_T^U - \lambda_U o_{V,T}^U \theta_{T-1}^U. \quad (13)$$

Before the final period, each trader's reservation price also reflects expectations about future trading prices, so it depends on both groups' beliefs.

3.2 Agency-trading equilibrium

Let

$$\nu_t := \gamma_t^U / \gamma_t^I, \quad w_t^I := \frac{N_M}{N_M + 1} \frac{N_I \nu_t}{N_I \nu_t + N_U}, \quad w_t^U := \frac{N_M}{N_M + 1} \frac{N_U}{N_I \nu_t + N_U}. \quad (14)$$

Then the equilibrium bid and ask prices can be written as weighted averages of the informed and uninformed reservation prices, plus the oligopoly wedge:

$$A_t = w_t^I P_t^{IR} + (1 - w_t^I) P_t^{UR} + \frac{\Delta_t^+}{N_M + 1}, \quad (15)$$

$$B_t = w_t^I P_t^{IR} + (1 - w_t^I) P_t^{UR} - \frac{\Delta_t^-}{N_M + 1}, \quad (16)$$

where $\Delta_t^+ = \max(\Delta_t, 0)$ and $\Delta_t^- = \max(-\Delta_t, 0)$.

The key implication is especially simple:

$$A_t - B_t = \frac{|\Delta_t|}{N_M + 1}. \quad (17)$$

Trading quantities and trading volume are

$$N_I(\theta_t^I - \theta_{t-1}^I) = -N_U(\theta_t^U - \theta_{t-1}^U) = \frac{N_M}{N_M + 1} \frac{\Delta_t}{\gamma_t^I / N_I + \gamma_t^U / N_U}, \quad (18)$$

$$\text{Vol}_t = \frac{N_M}{N_M + 1} \frac{|\Delta_t|}{\gamma_t^I / N_I + \gamma_t^U / N_U}. \quad (19)$$

Interpretation. In the agency case, the same reservation-price gap drives both **how much people want to trade** and **how much market makers can charge**. That is the paper's core explanation for why volume and spread can move together.

3.3 Why Figure 1 matters

Figure 1 in the paper draws the inverse demand and inverse supply schedules. When informed traders want to buy and uninformed traders want to sell, the ask price lies below the informed reservation price and the bid lies above the uninformed reservation price. The spread is exactly the piece of the surplus extracted by oligopolistic market makers.

Interpretation. The microstructure geometry is very close to a Cournot-style pricing problem. Market makers trade off price against quantity and earn profits from the wedge they create.

3.4 Principal-trading equilibrium

In the principal case, the market maker has its own reservation price:

$$P_t^{IR} = \bar{V} + \xi_t^I(\hat{V}_t^I - \mu_t X_t - \bar{V}) + \psi_t^I(\hat{V}_t^U - \bar{V}) - \gamma_t^I \theta_{t-1}^I, \quad (20)$$

$$P_t^{MR} = \bar{V} + \xi_t^M(\hat{V}_t^I - \mu_t X_t - \bar{V}) + \psi_t^M(\hat{V}_t^U - \bar{V}) - \gamma_t^M \theta_{t-1}^I, \quad (21)$$

$$P_t^{UR} = \hat{V}_t^U. \quad (22)$$

The quoted prices become

$$P_t^I = (1 - \omega_t^M)P_t^{IR} + \omega_t^M P_t^{MR}, \quad P_t^U = P_t^{UR}, \quad (23)$$

with $\omega_t^M > 0$. Trading quantities and volume are

$$\theta_t^I - \theta_{t-1}^I = \frac{\omega_t^M}{\gamma_t^I} (P_t^{IR} - P_t^{MR}), \quad (24)$$

$$\text{Vol}_t = \frac{N_I \omega_t^M}{\gamma_t^I} |P_t^{IR} - P_t^{MR}|. \quad (25)$$

At the last period,

$$A_T - B_T = \frac{1}{N_M + 1} |P_T^{IR} - P_T^{UR}|, \quad (26)$$

but for $t < T$ the spread depends on a weighted combination of the informed–uninformed gap and the market maker–uninformed gap.

Interpretation. Once market makers carry inventory and choose it dynamically, their own valuation enters the price system. That is why the principal case does not reduce to a simple spread formula based only on Δ_t .

3.5 Information-driven versus liquidity-driven markets

The paper first separates the two forces that can generate trading.

Information-driven market. When new private information arrives every period but liquidity shocks arrive only at the opening, both expected trading volume and spread decline over time.

- Uninformed investors and market makers learn gradually from prices.
- The difference between reservation prices shrinks.
- Both volume and the spread therefore get smaller as the day progresses.

Liquidity-driven market. When liquidity shocks arrive every period but new information arrives only at the opening, the spread tends to rise over time.

- Additional hedging shocks widen the gap in reservation prices.
- In the agency case, traders shift some hedging trades earlier to avoid very wide end-of-day spreads, so volume becomes hump-shaped before the close.
- In the principal case, volume rises more steadily toward the close because market makers smooth

the rise in spreads by managing inventory over the whole day.

Interpretation. Information alone gives a downward intraday pattern; liquidity shocks alone give a rising spread pattern. The double U-shape comes from combining them.

3.6 How the double U-shape emerges

When information and liquidity shocks both arrive at constant rates, traders behave asymmetrically:

- they **trade early on private information**, because delay means giving up informational rents if others also act on the signal;
- they **delay hedging trades**, because adverse selection becomes milder as more information is revealed through time.

This produces:

- high volume at the open from information trading,
- high volume near the close from delayed hedging,
- low volume in the middle,
- and, because market makers widen quotes when trading incentives are strong, a matching U-shaped bid-ask spread.

Interpretation. The model's main innovation is not merely that traders have two motives. It is that the **timing** of those motives is different, and market makers exploit that timing through their market power.

3.7 What makes the U-shape stronger or weaker

The model delivers a rich set of comparative statics.

Higher uncertainty of liquidity shocks. When $\tau_{\eta,t}$ is lower, hedging demand is more volatile. Uninformed traders then interpret order flow as more likely to be liquidity-driven, so adverse selection is weaker. Informed traders trade more aggressively at the open, and hedging demand is pushed more strongly to the close. The U-shape of volume becomes more pronounced, and the U-shape of the spread also becomes stronger.

Lower precision of private information. When $\tau_{\varepsilon,t}$ is lower, informed traders' signals are noisier. This again makes uninformed traders more willing to view trades as hedging-driven, which strengthens the U-shape in volume. By contrast, when private information becomes more precise, the opening spread widens because the informed reservation-price advantage at the open becomes larger.

More market makers or more trading rounds. Increasing N_M or increasing the number of within-day trading rounds reduces market power and narrows spreads throughout the day. The U-shape becomes less pronounced.

Interpretation. The comparative statics line up tightly with the core mechanism: whenever information problems are softer or market makers are more competitive, the extreme open/close pricing pattern weakens.

4 Calibration, applicability, and limits

4.1 Calibration design

The paper calibrates the model to match intraday NASDAQ patterns using TAQ data from 2000 to 2021. The calibration targets are:

- **14 half-hour absolute spread points** from 9:30 to 16:00;
- **13 half-hour turnover points** from 10:00 to 16:00.

To keep the problem tractable, the authors set

$$\bar{V} = 1, \quad \tau_V = 1, \quad \bar{\theta} = 1, \quad T = 13, \quad (27)$$

and assume $N_M = 1$ and $N_I = 1.5$. They then estimate six remaining parameters:

$$N_U, \lambda_I, \tau_{\varepsilon,0}, \tau_{\varepsilon,t}, \tau_{\eta,T}, \tau_{\eta,t}. \quad (28)$$

Interpretation. The calibration is not a full structural estimation of micro data. It is a disciplined attempt to see whether the mechanism can reproduce the **joint intraday shape** of spreads and volume.

4.2 Calibrated parameters

Parameter	Agency MM	Principal MM
N_U	283.480	N/A
λ_I	5.373	2.227
$\tau_{\varepsilon,0}$	0.238	0.168
$\tau_{\varepsilon,t}$	0.096	0.106
$\tau_{\eta,T}$	15.411	9.070
$\tau_{\eta,t}$	111.258	39.552

Two patterns stand out.

- Private information is estimated to be **more precise at the open** than later in the day.
- Hedging shocks are estimated to be **more important near the close** than earlier in the day.

Interpretation. The calibration backs the model's core verbal story: the open is information-heavy, the close is hedging-heavy.

4.3 What the calibration gets right

The calibrated model tracks the NASDAQ patterns fairly well.

- In the agency case, Figure 8 shows that the model reproduces both the large opening spread and the rebound near the close, as well as the U-shaped turnover path.

- In the principal case, Figure 9 shows a similar qualitative fit, with somewhat smoother volume dynamics.

Interpretation. The model is not meant to hit every point exactly. Its success is that a **single mechanism** can generate the main shapes in both objects simultaneously.

4.4 Where the model is most relevant

The paper is explicit that its results apply best to markets with **imperfect competition among liquidity suppliers**. The authors discuss:

- NASDAQ, where market-maker competition remained imperfect even after electronic trading expanded;
- hybrid exchange settings where designated market makers still matter;
- OTC markets where designated dealers quote both sides and often have substantial market power.

Interpretation. The theory is not a universal explanation for every market. It is a theory for markets where dealer power is still economically meaningful.

4.5 Important limitations

The paper makes several tractability-driven simplifications.

- It does **not** solve the fully dynamic case where risk-averse market makers simultaneously perform both agency and principal trading.
- It does **not** model a dynamic limit-order book with strategic customer limit orders.
- In the principal case, uninformed traders are taken to be risk neutral to keep the model closed form.
- The calibration is illustrative; it is not a formal empirical test with statistical fit criteria across all moments.

Interpretation. These are real limitations, but they are the price of obtaining a tractable dynamic model with endogenous spreads, depths, and trading volume.

4.6 Execution-cost extension

The last extension in the paper studies a one-period setting with exogenous market-maker inventory shocks. When market makers hold a large positive inventory, the ask moves closer to the competitive price; when they hold a large short position, the bid moves closer to the competitive price.

Interpretation. Traders who want to trade on the same side as an already inventory-loaded market maker face worse execution costs. This connects the model to recent evidence on designated-market-maker inventory shocks.

5 Tables and figures

5.1 Figure 1: inverse demand/supply and bid-ask spread

Read:

- The figure draws the informed reservation price at the top, the uninformed reservation price at the bottom, and the equilibrium ask and bid in between.
- The vertical gap between reservation prices is the fundamental disagreement in the market.
- Oligopolistic market makers compress quantity relative to the competitive benchmark and extract a spread equal to a fraction of that disagreement.

Economic message. The paper's most basic result is visible in one picture: volume and the spread are both driven by the same reservation-price gap. This is why they can move together.

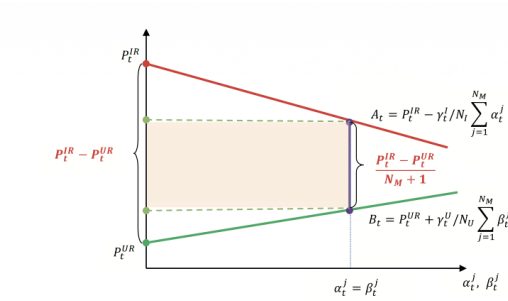


Figure 1
Inverse demand/supply functions and bid/ask spreads

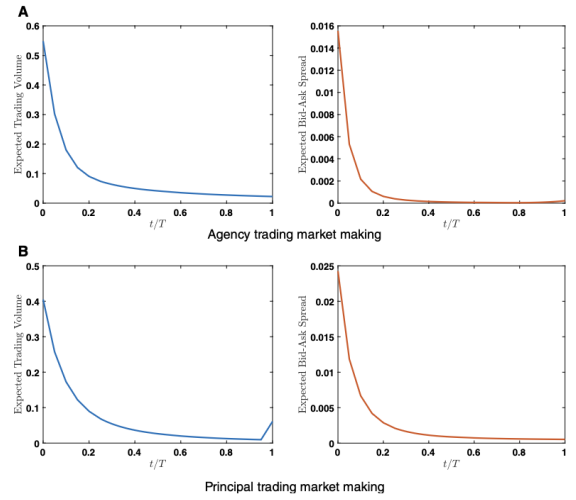


Figure 2
The dynamics of expected trading volume and spread against trading duration in information-driven markets

5.2 Figure 2: information-driven market

Read:

- With repeated information arrivals and no new liquidity shocks after the open, both expected volume and expected bid-ask spreads fall over the day.
- The decline is sharp at the open and then flattens, with a small end-of-day uptick in some cases because the close is the final trading opportunity.
- The principal case behaves similarly to the agency case here.

Economic message. Information alone does *not* generate the double U-shape. Instead it generates declining volume and declining spreads because the market gradually learns.

5.3 Figure 3: liquidity-driven market

Read:

- With only new liquidity shocks after the open, the spread rises over time because the disagreement created by hedging demand grows.
- In the agency case, volume becomes hump-shaped before the close because traders optimally move some trades forward.
- In the principal case, volume rises strongly into the close instead of displaying a hump.

Economic message. The distinction between agency and principal trading market makers matters most when hedging shocks arrive over time.

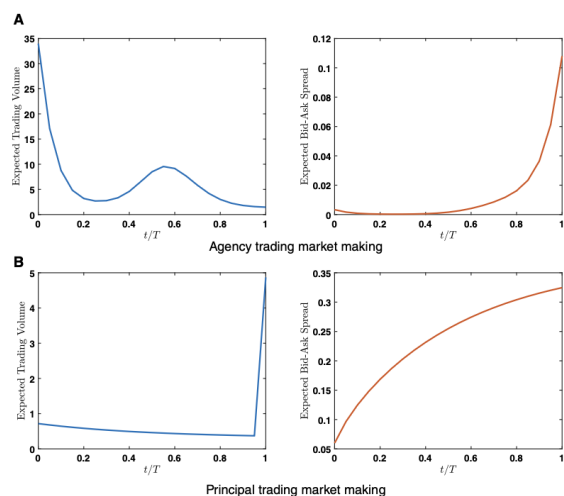


Figure 3
The dynamics of expected trading volume and spread against trading duration in liquidity-driven markets

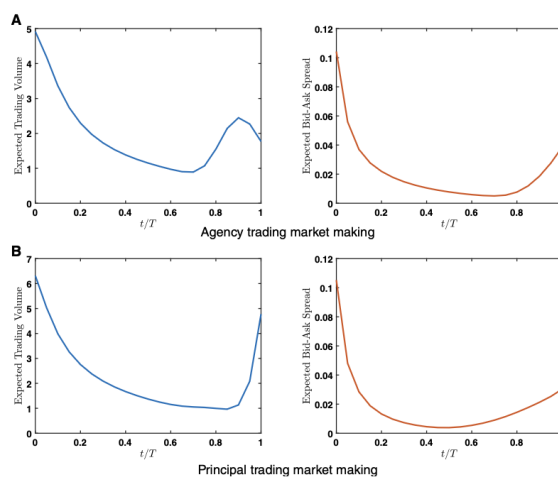


Figure 4
The dynamics of expected trading volume and spread against trading duration with both information and liquidity shocks arriving at constant rates

5.4 Figure 4: both information and liquidity shocks at constant rates

Read:

- Both expected volume and expected bid-ask spread are high at the open, low in the middle, and high again near the close.
- The left panels show the U-shape in volume; the right panels show the matching U-shape in spreads.
- In the agency case, the late-day peak can occur slightly before the close; in the principal case, it is pushed closer to the close itself.

Economic message. This is the paper's signature figure: the joint U-shape is an endogenous consequence of early information trading, late hedging, and market-maker power.

5.5 Figures 5 and 6: uncertainty and signal precision

Read:

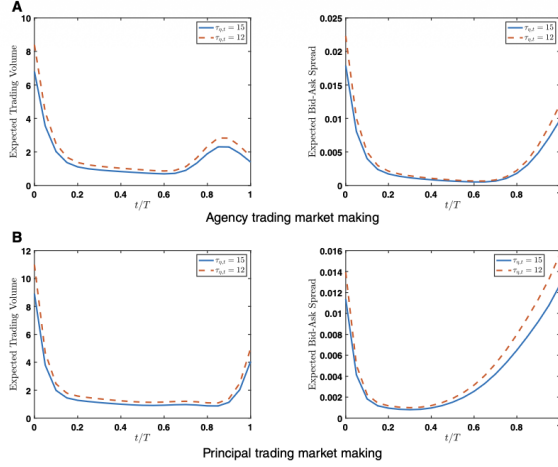


Figure 5
The dynamics of expected trading volume and spread against trading duration for different $\tau_{\eta,t}$

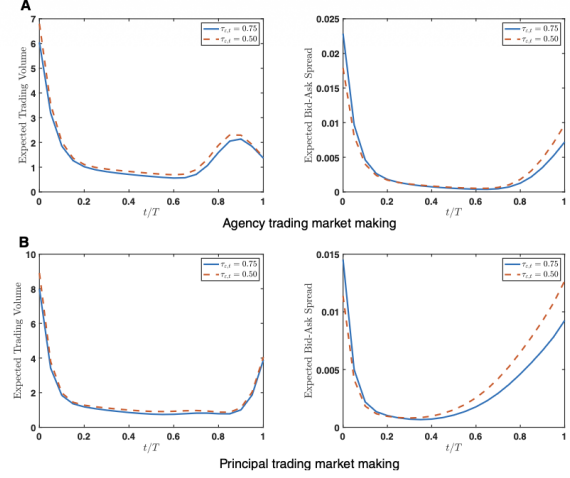
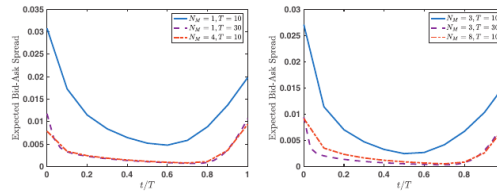


Figure 6
The dynamics of expected trading volume and spread against trading duration for different $\tau_{\epsilon,t}$

- Figure 5 shows that when liquidity shocks are more uncertain, the U-shape in both volume and spread becomes more pronounced.
- Figure 6 shows that noisier private information strengthens the U-shape in volume, while more precise information widens the opening spread because informed traders act more aggressively at the start of the day.

Economic message. The model predicts stronger open/close spikes when either hedging demand matters more or the market is more willing to interpret trades as hedging rather than private-information based.

5.6 Figure 7: market power and trading rounds



Read:

- Adding more market makers narrows spreads throughout the day.
- Increasing the number of trading rounds also compresses spreads.
- The U-shape is most pronounced when market makers are few and trading opportunities are sparse.

Economic message. This figure ties the model tightly to competition. Dealer market power is not a side detail; it is one of the main reasons spreads widen so much at the open and close.

5.7 Tables 1–3 and Figures 8–10: calibration and inventory shocks

Table 1
Intraday relative and absolute bid-ask spreads of NASDAQ stocks

Time	Absolute spread		Relative spread (%)		N
	Mean (1)	Std (2)	Mean (3)	Std (4)	
9:30	0.552	0.312	2.59	0.64	5,535
10:00	0.166	0.058	1.14	0.35	5,536
10:30	0.131	0.043	0.95	0.35	5,535
11:00	0.119	0.039	0.88	0.35	5,536
11:30	0.112	0.037	0.84	0.35	5,536
12:00	0.108	0.037	0.81	0.35	5,536
12:30	0.104	0.035	0.79	0.34	5,535
13:00	0.105	0.043	0.78	0.35	5,534
13:30	0.112	0.137	0.78	0.42	5,534
14:00	0.112	0.138	0.78	0.42	5,535
14:30	0.107	0.112	0.77	0.39	5,533
15:00	0.106	0.121	0.78	0.43	5,535
15:30	0.104	0.108	0.78	0.42	5,533
16:00	0.299	0.183	1.42	0.52	5,521

This table reports the average and standard deviation of bid-ask spreads at half-hour intervals throughout the trading day, specifically at 9:30, 10:00, 10:30, 11:00, ..., 15:00, 15:30, 16:00. At each of these trading points, we identify the bid and ask quotes that were placed nearest to the half-hour mark within a ± 5 minutes interval. For example, the quotes at 9:30 are taken as the quotes that were placed closest to 9:30 within the range of [9:25:00, 9:35:00]. We also require that the quotes must be submitted from the listed exchange (e.g., NASDAQ). We trim the relative spreads at 10% and absolute spreads at \$10 to avoid the influence of extreme outliers. We average bid-ask spreads across stocks and then across dates, generating average spreads at these 14 trading points. We also report the number of dates (N) at each time point available from January 2000 to December 2021.

Table 2
Intraday turnover of NASDAQ stocks

Time	Turnover (%)	Time	Turnover (%)
9:30–10:00	25.13	10:00–10:30	17.73
10:30–11:00	15.17	11:00–11:30	13.70
11:30–12:00	12.14	12:00–12:30	11.18
12:30–13:00	10.64	13:00–13:30	10.26
13:30–14:00	10.05	14:00–14:30	10.54
14:30–15:00	10.70	15:00–15:30	12.17
15:30–16:00	18.79		

This table reports the average turnover over half-hour intervals throughout the trading day, specifically during the time intervals of 9:30–10:00, 10:00–10:30, 10:30–11:00, ..., 15:00–15:30, and 15:30–16:00. We calculate the average turnover across stocks and then across dates from January 2000 to December 2021, generating average turnover at these 13 trading points. To avoid the influence of extreme outliers, we trim turnover at 200%.

Table 3
List of given and estimated parameters

Parameter	Description	Value	
		Agency MM	Principal MM
$\bar{V}$	Expected payoff	1	1
τ_V	Precision of payoff V	1	1
$\bar{\theta}$	Total supply	1	1
T	Trading periods	13	13
λ^U	Risk aversion of uninformed	1	0
N_M	Number of market makers	1	1
N_I	Mass of informed traders	1.5	1.5
N_U	Mass of uninformed traders	283.480	N/A
λ^I	Risk aversion of informed	5.373	2.227
$\tau_{e,0}$	Precision of information at period 0	0.238	0.168
$\tau_{e,t}$	Precision of information at other periods	0.096	0.106
$\tau_{\eta,T}$	Precision of hedging at T	15.411	9.070
$\tau_{\eta,t}$	Precision of hedging at other periods	111.258	39.552

The table reports the given parameter values for $\bar{V}$, τ_V , $\bar{\theta}$, T , λ^U , N_M , and N_I . The bottom six lines report the estimated parameters N_U , λ^I , $\tau_{e,0}$, $\tau_{e,t}$ for $0 < t \leq T$, $\tau_{\eta,T}$, and $\tau_{\eta,t}$ for $0 \leq t < T$.

Read:

- Table 1 shows the empirical U-shape in absolute and relative spreads for NASDAQ stocks.
- Table 2 shows the empirical U-shape in turnover.
- Table 3 reports calibrated parameters that make the open more information-driven and the close more hedging-driven.
- Figures 8 and 9 show that the model's predicted spreads and turnover align fairly closely with the empirical dots for both agency and principal market-making versions.

Economic message. The calibration does not merely fit one curve. It fits the joint behavior of spreads and turnover, which is the main empirical challenge the paper set out to solve.

Read:

- A large positive dealer inventory makes the ask close to the competitive price and leaves the bid farther away.
- A large negative inventory does the opposite: the bid is close to competitive, while the ask becomes much worse.

Economic message. Inventory shocks shift execution costs asymmetrically across buyers and sellers. This extension connects the paper's dynamic spread-setting logic to evidence on designated market makers' inventory positions.

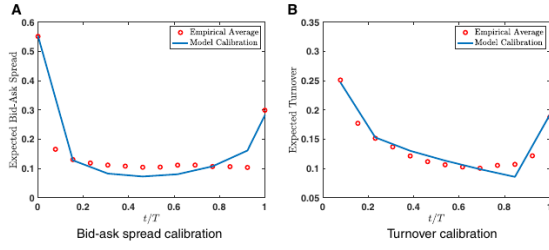


Figure 8
Model calibration of the average absolute bid-ask spread (left) and cumulative turnover (right) for stocks listed on NASDAQ for agency trading market makers

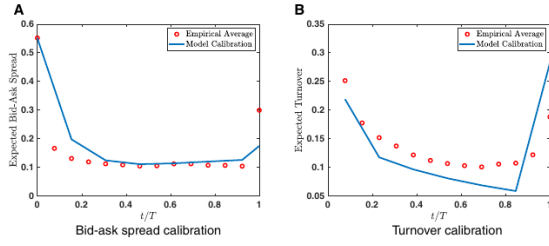
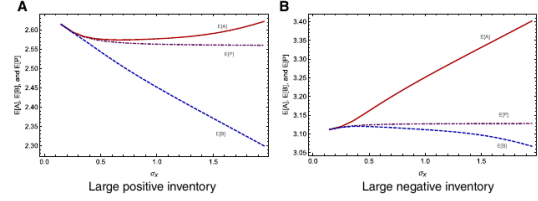


Figure 9
Model calibration of the average absolute bid-ask spread (left) and cumulative turnover (right) for stocks listed on NASDAQ for principal trading market makers



6 Short summary

- The paper explains the double U-shaped intraday patterns of trading volume and bid-ask spreads with a dynamic model of private information, liquidity shocks, and oligopolistic market makers.
- Informed traders trade early on information, while hedgers wait because adverse selection falls over time.
- Agency market makers make spread and volume both proportional to the reservation-price gap between informed and uninformed traders.
- Principal market makers add an inventory-management motive, so their own reservation price enters equilibrium pricing.
- The model predicts stronger U-shapes when market makers have more market power, when order imbalance is larger, and when hedging demand is more important.
- Calibration to NASDAQ TAQ data from 2000–2021 lines up well with the observed intraday spread and turnover curves.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that one can endogenously generate the observed intraday U-shapes in both trading volume and bid-ask spreads by combining private information, hedging demand, and imperfect competition among market makers in a multiperiod model.

7.2 Economic intuition

The open is dominated by information-based trading, because informed traders want to move quickly before their edge dissipates. The close is dominated by hedging, because traders are willing to wait until adverse selection is lower. Market makers exploit both moments by widening spreads when trading incentives are strongest.

7.3 Takeaway

Dynamic Market Making with Asymmetric Information and Market Power gives a clean microstructure explanation for why activity and trading costs peak at both ends of the day. The central insight is that dynamic order timing by traders and dynamic pricing power by market makers are inseparable if one wants to understand intraday market patterns.